

The Mechanism of Plastic Deformation of Crystals.
Part I.—Theoretical.

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Experiments on the plastic deformation of single crystals, of metals and of rock salt have given results which differ in detail but possess certain common characteristics.

In general the deformation of a single crystal in tension or compression consists of a shear strain in which sheets of the crystal parallel to a crystal plane slip over one another, the direction of motion being some simple crystallographic axis. The measure of this strain, which will be represented by s , is the ratio of the relative lateral movement of two parallel planes of slip to the distance between them. Thus it is defined in the same way as the shear strain considered in the theory of elasticity.

The resistance to shear, which will be denoted by S , is defined as the component of shear stress in the direction of slip which must act parallel to the slip plane in order that plastic deformation may occur.

It has been found that when the results of tests on single crystals of a metal are analysed the stress-strain curve which represents S as a function of s is independent of the stress normal to the slip plane and of the components of shear stress perpendicular to the direction of slip. Thus the (S, s) curve is a unique curve which defines the strength of the single crystal at any stage of distortion.

When the (S, s) curves for single crystals of different metals are compared they are found to differ considerably in detail, but they all possess one general characteristic, a very small stress will produce a small plastic deformation, but as the deformation increases the stress necessary to increase it also increases. With some crystals* it has been found difficult to assign a definite stress at which plastic distortion begins, with others, such as rock salt and zinc, experimenters have found such a limit, but it is very small compared with the strength ultimately attained by the material. In some such crystals it has been found that the observed lower limit of strength depends very much on the degree of purity of the material, thus Schmid,† working with single crystals of

* *E.g.*, aluminium.

† 'Z. Physik,' vol. 75, p. 538 (1932).

zinc, found that a decrease in the total amount of metallic impurity from 0.03% to 0.002% causes the stress at which the plastic deformation begins to decrease from 94 to 49 gm. per sq. mm.

Except for these differences in the early stages of distortion the (S, s) curves for many metallic single crystals are very similar. Some of them are shown in figs. 1, 2, and 3. Fig. 1 refers to aluminium. The data from which this curve

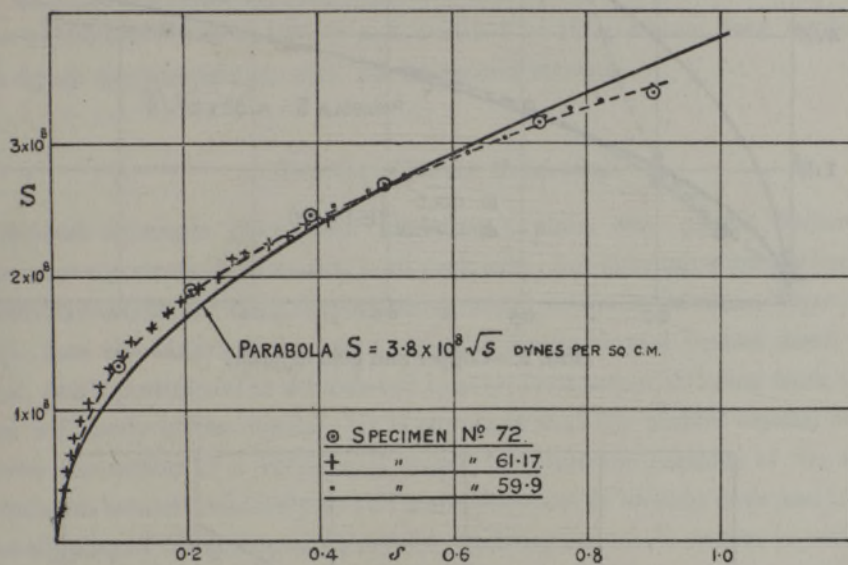


FIG. 1.—Aluminium crystals.

has been constructed were given in a previous paper.* In fig. 1, however, the unit of stress has been changed from lbs. per square inch to dynes per square centimetre. Fig. 2 shows the result of tests on single crystals of copper and gold.† These tests were made by Dr. Elam and published in the form of curves giving S for various extensions of the specimens. The additional data necessary for calculating s were kindly furnished by Dr. Elam, and the results set forth in fig. 2 were calculated by means of the formulæ previously given.‡

Fig. 3 shows the results of tests on four single crystals of iron. The crystals in question were compressed and the geometry of their distortion analysed and published in a previous paper.§ The stress data required for the (S, s) curve had been recorded, but were not included in the published results.

* Taylor, 'Proc. Roy. Soc.,' A, vol. 116, p. 51 (1927).

† Elam, 'Proc. Roy. Soc.,' A, vol. 112, p. 289 (1926).

‡ 'Proc. Roy. Soc.,' A, vol. 116, p. 47 (1927), equations (2), (4), (5), (6).

§ Taylor, 'Proc. Roy. Soc.,' A, vol. 112, p. 338 (1926).

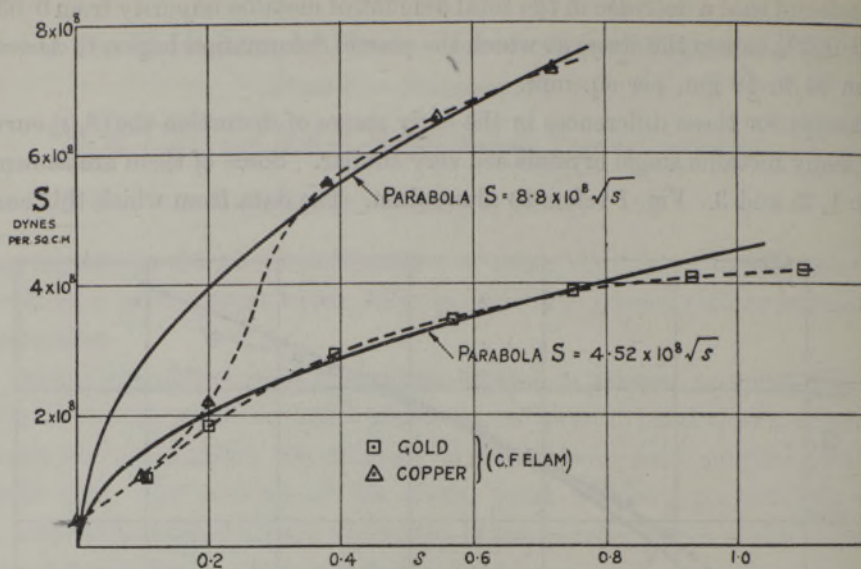


FIG. 2.—Copper and gold crystals.

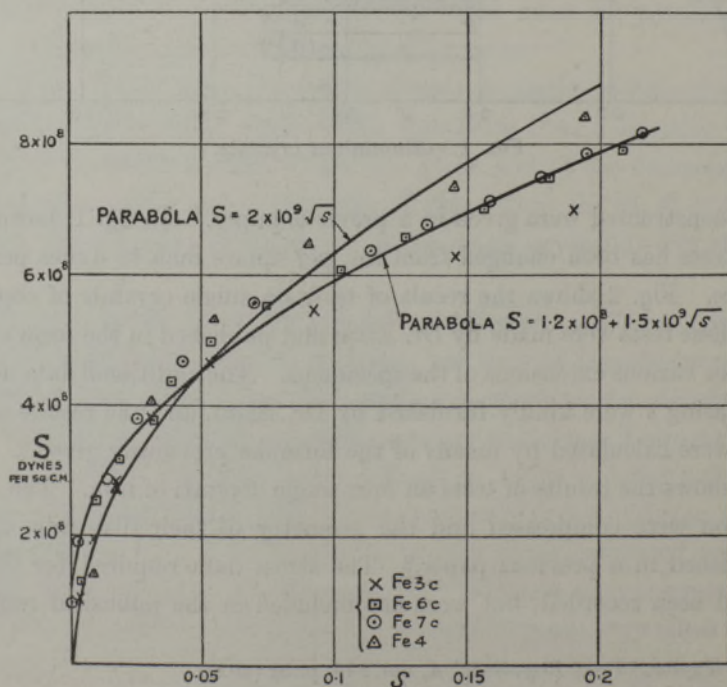


FIG. 3.—Iron crystals.

It will be seen that for iron and aluminium at any rate the curves resemble the parabolas.

Such parabolas have been drawn in figs. 1, 2, and 3 in order that they may be compared with the results of observation. For copper, fig. 2, the first three observed points do not lie on the parabola, while for iron, fig. 3, it seems that the experiments indicate a finite value (about equal 1.2×10^8 dynes per sq. cm.) for the stress at which plastic distortion first occurs. The parabolas $S = 1.2 \times 10^8 + 1.5 \times 10^9 \sqrt{s}$ and $S = 2.0 \times 10^9 \sqrt{s}$ have been marked on the figure for comparison with the measured stresses.

Theories of Strain Hardening.

Several attempts have been made to explain why plastic deformation increases the strength of metals (and rock salt), but they have mostly been of a qualitative nature. Such explanations are principally of three types. One type uses the observed fact that, when a single crystal breaks down under load, small portions of it are rotated into an orientation differing from that of the main body of the crystal. It is supposed that the perfect crystal can slip under the action of a very small stress, but that the portions of the crystal which are rotated relatively to the main body act as locking keys and hold up the slipping of the surrounding crystal, enabling the whole system to withstand a much greater load than the perfect crystal can support.

When the crystal contains *only* one set of planes of easy glide parallel to one crystallographic plane this theory might be capable, after some modification, of explaining a very rapid strengthening with increasing distortion. Where, however, the portions inside and outside a surface of misfit are supposed to be perfect crystals, but with crystallographic axes oriented in two different directions, easy gliding in the outer crystal would produce a definite strain (*i.e.*, alteration of shape) in the surface of misfit while easy gliding in the inner crystal would produce a different strain. Thus the only types of strain which are possible as a result of easy gliding in both the inner and the outer crystal are geometrically inconsistent at the boundary between them. If no other type of plastic strain is possible, then the inner crystal completely locks the portion of the outer crystal within which the planes of easy glide cut any portion of the inner crystal.

When the crystals are capable of slipping equally easily on several crystallographically similar planes, as, for instance, when the crystals have cubic symmetry, this explanation of hardening due to cold work is no longer applicable,

because, by an appropriate combination of easy glides, the inner crystal can be given exactly the same strain as the outer one. Under these conditions, therefore, there would be no geometrical inconsistency at the surface between them.

It is true that the stress necessary to cause any given kind of strain is likely to vary slightly according to whether the crystallographic axes are arranged in the most favourable or the most unfavourable orientation for the particular type of strain considered, but this effect is necessarily very small compared with the very large effect which the theory was designed to explain.

We may now turn to a second type of explanation. A perfect crystal is again supposed to be capable of withstanding only a very small stress, but every real crystal contains surfaces of misfit which form a mosaic structure or, as some crystallographers have described them, a system of lineages. These surfaces of misfit are supposed to hold up slipping to an increasing extent as the amount of distortion proceeds. This explanation, though at first sight promising, entails very great difficulty as soon as it is examined in detail.

The idea that the boundary can hold up slip in the interior of a perfect portion of the crystal might be interpreted in two ways. In one the surfaces of misfit may be regarded as being capable by themselves of carrying the whole applied stress, the intervening portions of perfect crystal structure being in a stress-free state. The strength would then be like that of a honey-comb, depending entirely on the boundaries which receive no assistance from the contents of each cell. Even if it were possible to explain the existence of any given strength on this principle it would be extremely difficult to explain why this increases with the amount of plastic strain.

In the other interpretation any thin sheet bounded by two parallel slip planes must move as a whole. Each perfect portion of the crystal can support any stress system for which there is no component of shear stress parallel to the slip planes, and the varying orientations of neighbouring portions which are crystallographically perfect might enable the whole system to support a large total stress in spite of the fact that each portion was excessively weak to one type of shear stress. This interpretation would make theory identical with the first type already discussed. It breaks down if the crystal has several crystallographically similar possible slip planes.

The third type of explanation is the inverse of the first two. A perfect crystal is supposed to be capable of withstanding a very large stress. The observed weakness of metal crystals is attributed to concentrations of stress due to internal surfaces of misfit or cracks, and the increasing strength with

increasing plastic strain is attributed to an increase in the number of faults or cracks.* As the number of such faults increases the ratio of maximum stress in a region of stress concentration to mean stress in the material would be expected to decrease. The mean stress necessary to cause a given maximum stress in a region of stress concentration must, therefore, increase as the number of faults per unit volume of the material increases.

In the present paper an attempt will be made to present a theory of this kind in a quantitative form. It will be found that the idea that a system of faulting exists in the structure of an apparently perfect crystal arises naturally in the course of the work.

Significance of Direction of Slip in Crystals.

Perhaps the most remarkable feature of the distortion phenomena of single crystals is that during the whole process of slipping parallel to one crystal plane they slip in a definite crystallographic direction. Regarding the matter from the atomic point of view the finite rigidity of the deformed crystal implies that each atom is situated in a position of stable equilibrium in the field of force due to its neighbours, and to intervening free electrons if there are any. Plastic strain must be due to stability interchanges in which atoms jump, owing to thermal or other agitation and to the applied stress, from one position of equilibrium to another. If these jumps are due to thermal agitation, they must be regarded as random occurrences though, since each jump alters the field of force in its neighbourhood, the probability that any given jump will occur may change very greatly when a jump occurs in a neighbouring position. From this point of view the whole macroscopic phenomenon of gliding must be regarded as the integrated effect of individual jumps, and since the direction of the resultant glide is parallel to a line of atoms in the crystal, it is difficult to avoid the conclusion that a great majority at any rate of the individual jumps are in this direction too.

If this is so, the length of the jump is likely to be equal to the spacing of atoms along this line, and the mechanism of slipping may be more like the simple shift shown in fig. 4 (c), in which the whole of the material on one side of a definite plane shifts through the length of one lattice cell, than has often been supposed. It is proposed, therefore, to examine how this ideal condition differs from what is observed in real materials. In the first place this ideal slipping would leave the material in the form of a perfect crystal and the

* Taylor, 'Trans. Faraday Soc.,' vol. 24, p. 121 (1928).

strength would be unaltered by the distortion. In the second place to shift the whole of the upper row of atoms simultaneously over the lower row would necessitate the application of a stress comparable with, though no doubt less than, the elastic moduli of the material. Further, it would leave no room for an explanation of the large observed effect of temperature on plastic distortion.

Slipping Over a Portion of the Slip Plane.

It seems that the whole situation is completely changed when the slipping is considered to occur not simultaneously over all atoms in the slip plane but over a limited region, which is propagated from side to side of the crystal in a finite time. It has been observed by Joffe* that the first sign of breakdown when a stressed crystal of rock salt is examined between nicol prisms is the appearance of a bright line, indicating distorted material, which is propagated along a slip line from side to side of the crystal. The stresses produced in the material by slipping over a portion of a plane are necessarily such as to give rise to increased stresses in the part of the plane near the edge of the region where slipping has already occurred, so that the propagation of slip is readily understandable and is analogous to the propagation of a crack.

If, in order to retain the explanation of the observed constancy of the direction of slip, we assume that the propagation of a line of slipping along a slip plane leaves a perfectly well ordered crystal arrangement in its wake, we obtain a system represented pictorially in fig. 4. In the diagram (a) represents the atoms in the lattice of a crystal block; (b) the condition of this block when a slip of one atomic spacing has been propagated from left to right into the middle; and (c) the block after the unit slip, or "dislocation" as we may call it, has passed through from left to right.

Positive and Negative Dislocations.

The configurations a, b, and c of fig. 4 represent the passage of a dislocation in which the atoms above the operative slip planes are compressed in the direction of slip while those below are expanded. It is clear that a similar dislocation can exist which is the mirror image in the slip planes of that shown in b, fig. 4. This configuration which is shown in fig. 4 (e) will be called a negative dislocation to distinguish it from fig. 4 (b) which will be called a positive dislocation. The passage of a positive dislocation across a crystal

* "The Physics of Crystals," p. 47.

block from left to right produces the same effect as the passage of a negative one from right to left, namely, the shift of the upper part of the block through one atomic spacing to the right relative to the lower part. This is illustrated in fig. 4 by the sequence of configurations *d*, *e*, *f*.

It seems likely that the motion of a "unit dislocation" may be determined by the stress and the temperature. It is known that in the case of metals there is a recrystallization temperature, below the melting point, at which large irregularities in structure such as those which must exist at the boundaries of crystal grains with different crystal orientations can be propagated through

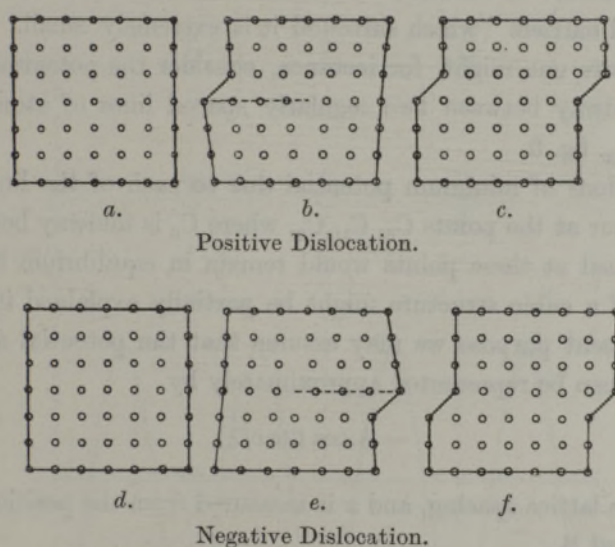


FIG. 4.—Positions of atoms during the passage of a dislocation.

the material. It may well happen therefore that there is a definite temperature probably lower than the recrystallization temperature, at which a unit dislocation of the type described above may be capable of free movement in either direction along the crystal plane. Let us call this hypothetical temperature T_D .

This picture seems adequate to account for the fact observed at high temperatures that slipping may occur at very low stresses in single crystals and that, after an indefinite amount of distortion, the crystal is still nearly in its original condition, except for change in shape; it is still a single crystal and still has very little strength. The distortion would then be accounted for by the propagation of centres of dislocation along the slip planes. These would enter the specimen at one side and leave it at the other as shown in fig. 4.

Atomic Model of a Dislocation.

Some insight into the manner in which a dislocation might be propagated along a slip plane may be gained by introducing a conception due to Dehlinger* to explain recrystallization and described by him as "Verhakung" or "hooking." In theories of the equilibrium of crystal lattices an atom is supposed to place itself in a position of minimum potential energy in relation to its neighbours. Heat motions will agitate the atom so that it moves about in the neighbourhood of this position of minimum potential energy, but, until a certain temperature is reached, the chance that the atom will escape across the "potential barriers" which surround it is extremely small. In a perfect crystal structure one might, for instance, consider the potential along a line CD spaced midway between two regularly spaced lines of atoms A_0, A_1, A_2 and B_0, B_1, B_2 , fig. 5.

If the positions of minimum potential due to each of the lines considered separately occur at the points C_0, C_1, C_2 , where C_0 is midway between A_0 and B_0 atoms placed at these points would remain in equilibrium there and the equilibrium of a cubic structure might be partially explained in this way.

For the present purpose we may assume that the potential along CD due to either row can be represented approximately by

$$-A \cos (2\pi x/\lambda), \quad (1)$$

where λ is the lattice spacing, and x is measured from the position C_0 midway between A_0 and B_0 .

The total potential due to both rows, namely, $A_0, A_1, A_2, \dots$ and $B_0, B_1, B_2, \dots$, is then

$$\phi = -2A \cos (2\pi x/\lambda). \quad (2)$$

The height of the potential hill or barrier which must be climbed by any atom in order to escape altogether from its equilibrium position is then $4A$. This is represented graphically by the sine curve of amplitude $2A$ which is shown in fig. 5.

In the neighbourhood of a positive dislocation the atoms in the slip plane above the centre of dislocation are compressed in the direction of slip while those in the plane below it are extended. At some distance from the centre the spacings along lines of atoms above and below the path of the centre of dislocation become regular and equal to the normal spacing for a perfect

* 'Ann. Physik,' vol. 2, p. 749 (1929).

crystal, but if the number of atoms in a length L of a line of atoms which passes below the centre of a unit dislocation is N , then the number in the same length L on a line which passes above the centre is $N + 1$.

To calculate the distribution of potential on the line of atoms along which the actual centre of dislocation passes it would be necessary to solve the very difficult problem of finding the equilibrium positions of all the surrounding atoms. Though this is not yet possible the general nature of the potential

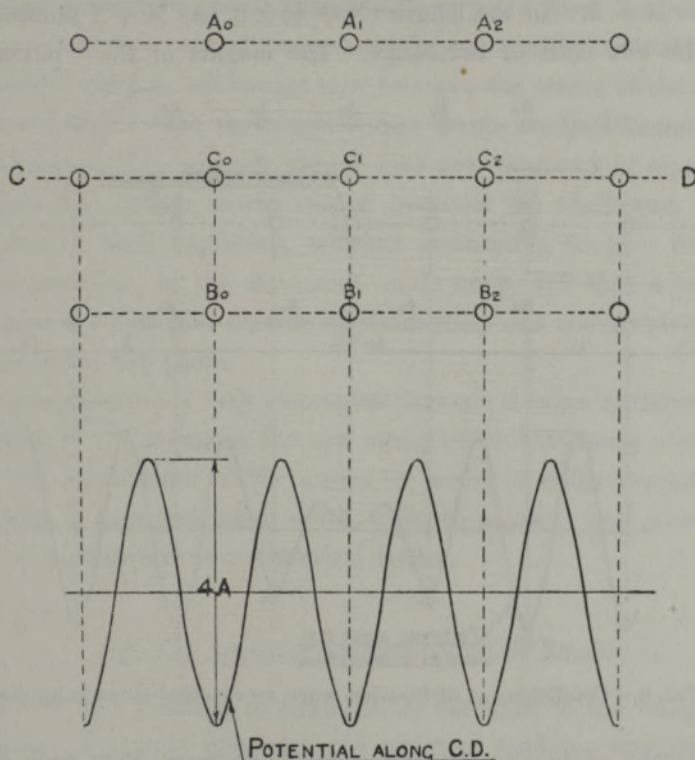


Fig. 5.—Arrangement of atoms in a perfect crystal structure.

distribution on the line along which the centre of the dislocation passes may perhaps be inferred from the results of calculation of potential due to an arbitrary distribution of atoms which has the essential characteristic that $N + 1$ atoms in a line above the centre correspond with N atoms below it.* The simplest arrangement of this kind is when $N + 2$ atoms in the upper line are evenly spaced over a length $(N + \frac{1}{2}) \lambda$ while the $N + 1$ atoms in the lower line are evenly spaced over the same length. This arrangement is shown in

* These numbers include the atoms at each end of the line so that $N + 1$ of the compressed atomic spacings in the upper row correspond with N spacings in the extended lower row.

fig. 6 for $N = 3$. Outside this range we may suppose the two rows to have the spacing λ of the unstrained crystal. Making the same approximation as that for a regular lattice, the potential along the line CD midway between these two lines is

$$P = -A \cos 2\pi \frac{x}{\lambda} \left(\frac{N+1}{N+\frac{1}{2}} \right) - A \cos 2\pi \frac{x}{\lambda} \left(\frac{N}{N+\frac{1}{2}} \right). \quad (3)$$

The nature of this function is well known. It is shown graphically in fig. 6 for the case $N = 3$. In the length $(N + \frac{1}{2})\lambda$ it has $N + 2$ minima including those at the two ends of the range. The heights of the "potential hills"

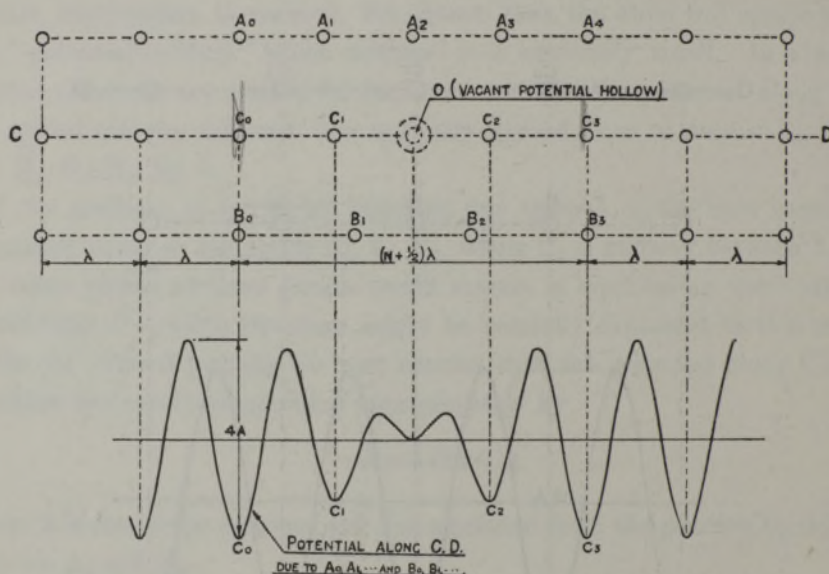


FIG. 6.—Conditions at dislocation when no external stress is applied.

and the depths of the "potential valleys" decrease from each end to the centre of the range. If N is an odd number, the shallowest depression on the potential curve is in the centre of the range. If we now imagine that each of the potential depressions except the central one is occupied by an atom we have a rough picture of a possible equilibrium distribution of atoms in the neighbourhood of a centre of dislocation. The actual centre of the dislocation may be regarded as being situated at the point O , fig. 6, in the middle of the central vacant potential hollow.

It has been pointed out that in a perfect crystal the atoms may be expected to remain within their potential barriers till a certain temperature is reached at which they can jump these barriers. At this temperature the atom C_0 , fig. 6, in the disturbed lattice might be able to jump into the space occupied

by C_1 . At lower temperatures an atom would not be capable of surmounting potential hills between C_0 and C_1 , but might still be able to jump the lower hills nearer the centre of the disturbance. The lowest temperature at which a jump can be made is that which will just permit either of the two central atoms to jump one of the lowest hills into the vacant central hollow O, fig. 6. This temperature we may call T_D . It may be noticed that even at the absolute zero of temperature some energy of agitation still persists in the crystal. If this is large enough to permit jumps to take place, then $T_D = 0^\circ$ abs.

When one of the two central atoms has jumped into the central hollow, the potential hollow which is left vacant now becomes the centre of the dislocation and we must imagine that the neighbouring atoms readjust themselves by a continuous process (*i.e.*, without jumps) into new positions of equilibrium so that the potential hollow newly vacant becomes the shallowest one. This idea has already been expressed, without attempting to give it a definite atomic interpretation, in the statement made on p. 369 that a temperature T_D exists above which the centres of dislocation can move freely in either direction along the slip plane.

The foregoing picture is very incomplete because it takes no account of the mutual action of the atoms in the row along which the jumps occur, but the theory of the equilibrium of the atoms in *perfect* metallic crystals is so incomplete that it does not seem worth while to attempt any more complete discussion of equilibrium in a disturbed lattice.

Effect of Applying an External Shear Stress.

We are now in a position to examine, in the light of the foregoing rough representation of atomic structure, the effect of applying an external shear stress. We may imagine that a shear stress applied in the direction indicated by the arrows in fig. 7 gives the row of atoms $A_0, A_1, A_2, \dots$, a small displacement to the right relative to the row $B_0, B_1, B_2, \dots$. If this small displacement is δ , the potential along the line CD is now

$$P = -A \cos 2\pi \frac{x - \delta}{\lambda} \left(\frac{N+1}{N + \frac{1}{2}} \right) - A \cos 2\pi \frac{x}{\lambda} \left(\frac{N}{N + \frac{1}{2}} \right). \quad (4)$$

It will readily be seen that the effect of this is to lower the height of the hill which the atoms to the right of the central hollow must jump in order to enter the central hollow. The hill which the atoms to the left of the centre must jump is correspondingly raised. The actual potential curve for $N = 3$,

$\delta/\lambda = 0.1$ is shown in fig. 7. The positions of the atoms and of the vacant potential hollow or centre of the dislocation are also shown in fig. 7. Here height of the potential hill h_1 , fig. 7, which the atom C_1 to the left of the centre of dislocation has to jump in order to land in the central potential hollow is $2.70 \text{ \AA}$. The atom C_2 to the right of the centre has only to jump a hill of height $h_2 = 0.85 \text{ \AA}$.

If the temperature is high enough to permit the jump h_2 but not high enough to permit the jump h_1 , the only jump which can occur is to the left into the centre of dislocation from the potential hollow on its right. The actual centre, *i.e.*, the vacant hollow, can therefore only move to the right. At higher

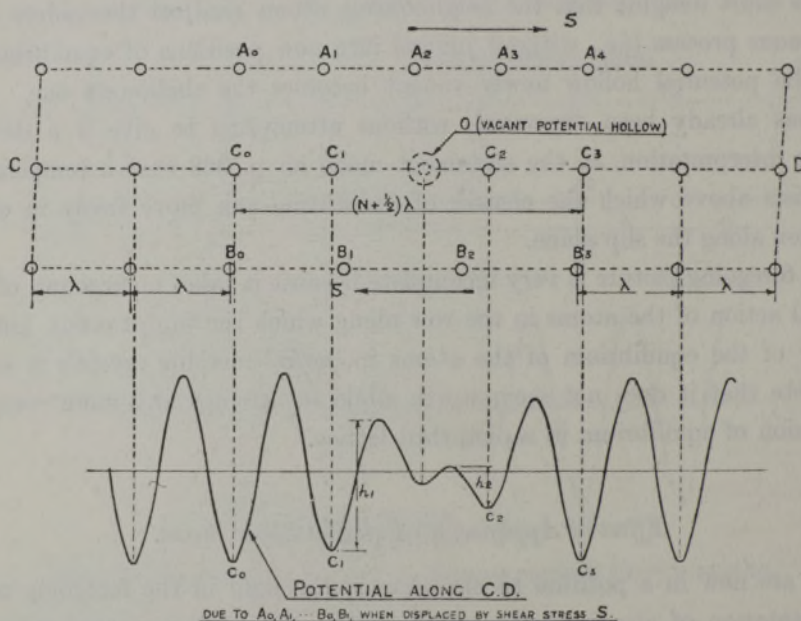


FIG. 7.—Conditions at dislocation in crystal subjected to shear stress.

temperatures, when the jumps h_2 and h_1 can both occur, the former will be the more frequent so that a positive centre of dislocation will necessarily migrate to the right under the action of a shear stress directed to the right. Similarly a negative dislocation will move to the left under the action of the same shear stress.

When the temperature is high enough to permit free movement of the centre of dislocation in the absence of applied shear stress the application of the smallest shear stress will cause the centre of dislocation to migrate along the slip plane. Thus plastic strain will occur when the smallest stress is applied. At lower temperatures a finite shear stress may be necessary in

order that the potential hill on one side of the centre of dislocation may be reduced in height to such an extent that the neighbouring atom can jump into the central hollow.

In the foregoing discussion an atomic mechanism is roughly represented which would give metallic crystals the following properties :—

- (1) At temperatures above T_D any stress, however small, applied parallel to the slip plane in the positive direction (see fig. 7) will cause positive centres of dislocation to travel in the positive direction and negative centres to travel in the negative direction ;
- (2) At temperatures below T_D the centres will not move till the shear stress attains some finite value.

In the work which follows these properties will be assumed and some possible consequences of their existence will be explored. It must be remembered that (1) is an accurate statement of what is to be expected of the atomic mechanism, but (2) is only a rough representation of some more complicated relationship involving temperature and rate of deformation which would result from a more complete consideration, from the statistical point of view, of the connection between temperature and the frequency of the stability interchanges, to which the motion of a dislocation is here ascribed.

Distribution of Stress near a Unit Dislocation.

The distribution of stress due to a unit dislocation cannot be calculated at points within a few lattice spacings of the centre of the disturbance, but at greater distances the theory of elasticity must apply. The general theory of dislocations in continuous isotropic elastic solids has been treated extensively by Volterra.* In the neighbourhood of a centre of dislocation the stresses become very large so that it is necessary to imagine that the material is cut away round the actual centre. We may therefore suppose that our material is initially unstrained and that it contains a hole H, fig. 8. From some point A on the boundary a centre of dislocation passes along a straight line AB to the point B which is on the surface of the hole H.

If P is a particle just above AB and P' is just below it, they suffer a relative displacement λ during the passage of the centre of dislocation between them.

The appearance of a circular ring which has suffered this kind of strain is shown in fig. 8. The stress distribution round such a dislocation has been

* 'Ann. Éc. Norm.,' vol. 24 (1907).

calculated by Timpe.* It depends on the surface tractions applied at the surface of the hole H as well as on the amount of slip; but if the resultant force and couple due to these surface tractions are both zero the stresses due to the dislocation vary as r^{-1} , r being the distance from the centre O, while the stresses due to the surface tractions vary as r^{-2} or higher inverse powers.

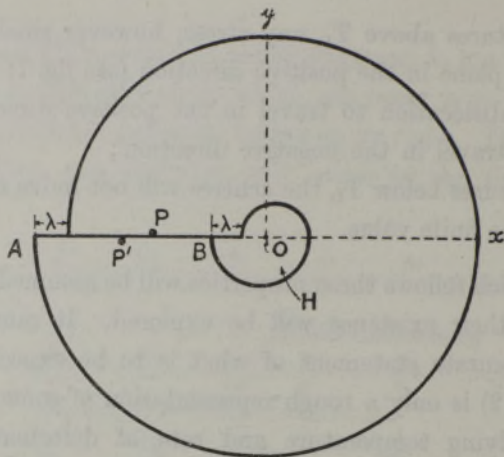


FIG. 8.—Elastic cylinder with dislocation.

At points far from the centre the components of stress due to a positive dislocation or slip of amount λ on the positive axis of x are

$$X_x = -Y_y = -\frac{\mu\lambda}{\pi} \frac{y}{x^2 + y^2}, \quad X_y = \frac{\mu\lambda}{\pi} \frac{x}{x^2 + y^2} \quad (5)$$

or in polar co-ordinates

$$\widehat{r}r = -\widehat{\theta}\theta = +\frac{\mu\lambda \sin \theta}{\pi r}, \quad \widehat{r}\theta = \frac{\mu\lambda \cos \theta}{\pi r}, \quad (6)$$

where μ is the modulus of rigidity.

In the particular instance when no traction is exerted on the surface of a circular hole H of radius a , the complete expressions for the components are

$$\left. \begin{aligned} \widehat{r}r &= +\frac{\mu\lambda}{\pi r^3} \sin \theta (r^2 - a^2) \\ \widehat{r}\theta &= \frac{\mu\lambda \cos \theta}{\pi r^3} (r^2 - a^2) \\ \widehat{\theta}\theta &= -\frac{\mu\lambda}{\pi r^3} \sin \theta (r^2 + a^2) \end{aligned} \right\} \quad (7)$$

These are identical with (6) when a is small compared with r .

* 'Göttingen Diss.,' Leipzig (1905).

It appears that whatever may be the state of stress at distances from the dislocation centre comparable with λ the stress at large distances is represented by (5). Thus, even though nothing is known about the exact conditions of equilibrium of atoms at a dislocation, the stress distribution at a distance of several lattice cells away is known, for it depends only on the amount of the slip. The application of the theory of dislocations in an elastic body to a crystal is, strictly speaking, only justifiable if the crystal is elastically isotropic, a condition which appears to be fulfilled for aluminium. In cubic crystals the error due to want of isotropy is likely to be small. It will be left out of consideration in the following pages.

Mutual Action and Equilibrium of Centres of Dislocation.

We may consider a single set of parallel planes as possible slip planes. If these are taken as perpendicular to the axis of y , the application of a positive shear stress will cause a positive centre to move to the right or a negative centre to the left. For the present we may suppose that the temperature is above T_D so that the dislocation moves, however small X_y may be. Since each centre of dislocation produces a distribution of shear stress in its neighbourhood, every centre must tend to cause all neighbouring centres to migrate along their slip planes.

Referring to equation (5) it will be seen that in the neighbourhood of a positive centre of dislocation, A , X_y is positive when x is positive, if therefore there is another positive centre B whose x co-ordinate relative to A is positive, B will move to the right under the influence of A . That is, positive centres of dislocation repel one another. Similarly two negative centres repel one another. On the other hand, positive centres attract negative centres and *vice versa*. If there are only two centres in the whole field and they are both of the same sign, they will repel one another to an infinite distance (provided the temperature is above T_D). Centres of opposite sign will attract one another till the line joining them is perpendicular to the slip plane, when they will rest in equilibrium under their mutual influence. This position is illustrated in fig. 9 (a).

If two centres of opposite sign lie on the same slip plane, they will attract one another, but, since the elastic equations cannot be applied when they come within a few lattice distances of one another, it is not possible to say whether or not they will neutralize one another and leave a perfect crystal structure.

We may now consider the effect of applying a positive shear stress S so that the upper face of a crystal block is subjected to a tangential stress acting towards the right. If only one dislocation is present, it will move across the block, to the right if it is positive and to the left if it is negative, however small S may be. In either case the result is the same, namely, a shift of the upper portion of the block relative to the lower part through a distance λ . Apart from this shift the properties of the block are unchanged by the passage of the dislocation.

Conditions are very different when two dislocations of opposite sign are present. Suppose, for instance, that owing to the application of a very small

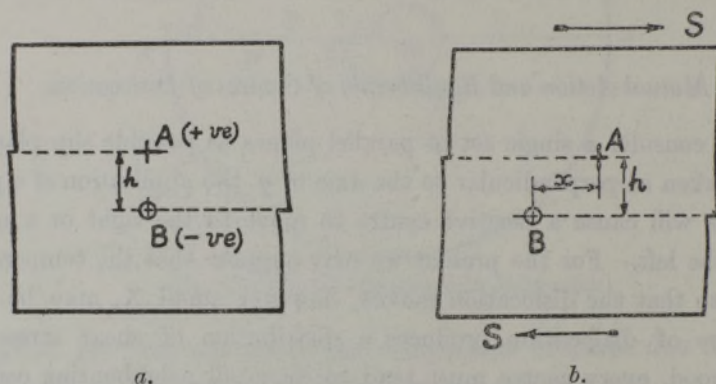


FIG. 9.—Positive and negative dislocations in equilibrium. (a) Under no external stress. (b) when shear stress is applied.

shearing force a positive dislocation starts moving along a slip plane from the left, and a negative dislocation starts to move from the right along another slip plane, distant h from the first. Even without application of the shear stress they would arrive under their mutual influence at the positions A, B, fig. 9 (a) where AB is perpendicular to the slip planes. On the other hand, in order to make them continue to move along their respective slip planes beyond AB, a shear stress must be applied. A shear stress S will cause the positive dislocation to move to the right and the negative one to the left till

$$S = \frac{\mu \lambda x}{\pi (x^2 + h^2)}, \quad (8)$$

where x is the projection on the slip plane of the distance between them. In this position each dislocation is in equilibrium.

The connection between S and x is shown in fig. 10. It will be seen that as x/h increases $S\pi h/\mu\lambda$ increases till the maximum value 0.5 is reached at $x = h$. The maximum possible value of S consistent with equilibrium is therefore

$\frac{1}{2} \frac{\mu\lambda}{\pi h}$ or $0.16 \mu\lambda/h$. If S is less than this, the two centres of dislocation cannot escape from one another, whereas when $S > \mu\lambda/2\pi h$ they pass one another and escape from their mutual attraction. Thus plastic distortion or slipping can only proceed, so far as the movement of these two centres of dislocation are concerned, when $S > \mu\lambda/2\pi h$. This is remarkable in view of the fact that the strength of the material is zero when only one centre of dislocation is present. Since S is inversely proportional to h it appears that slipping will

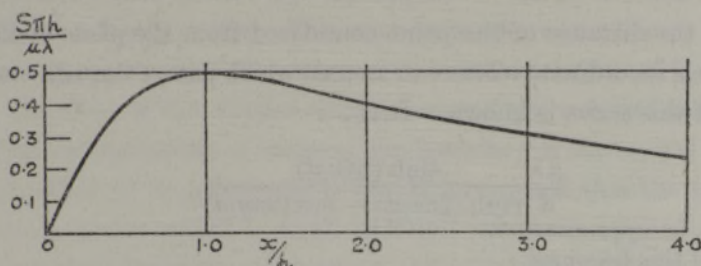


FIG. 10.

proceed on the slip planes which are farthest from planes of slip already containing centres of dislocation.

We may now consider possible equilibrium arrangements of centres of dislocation. It has been seen that positive and negative centres cannot exist on the same slip plane, for they would run together and neutralize one another. The simplest arrangements in which every centre is in equilibrium with its neighbours are those in which all the positive centres and all the negative

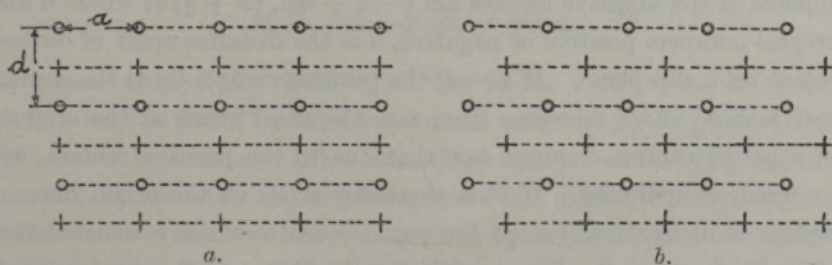


FIG. 11.—Equilibrium arrangements of dislocations, (a) stable, (b) unstable.

centres lie on interpenetrating rectangular nets. Two possible equilibrium configurations are shown as (a) and (b), fig. 11, where the positive centres are represented by crosses and the negative ones by round spots. The application of a positive shear stress to a crystal containing such a system would move all the positive centres to the right and all the negative ones to the left along the slip planes, represented by broken lines in fig. 11.

We shall calculate the displacement of the positive lattice relative to the negative when a given shear stress S is applied to the crystal. For this purpose we must first calculate the distribution of X_y due to the complete lattice of dislocations. The shear stress due to a row of positive dislocations regularly spaced at distance d apart in a line perpendicular to the slip planes is

$$\frac{\mu\lambda}{\pi} \sum_{n=-\infty}^{+\infty} \frac{x}{x^2 + (y + nd)^2} \quad (9)$$

where x is the distance of the point considered from the plane of the dislocations and y is its ordinate relative to an axis which passes through one of them. The sum of this series is known. It is

$$\frac{\mu\lambda}{d} \frac{\sinh(2\pi x/d)}{\cosh(2\pi x/d) - \cos(2\pi y/d)}; \quad (10)$$

when $y = 0$ this becomes

$$\mu\lambda d^{-1} \coth(\pi x/d); \quad (11)$$

when $y = \frac{1}{2}d$ it is

$$\mu\lambda d^{-1} \tanh(\pi x/d); \quad (12)$$

when $y = \frac{1}{2}d$ it is

$$\mu\lambda d^{-1} \tanh(\pi x/d). \quad (13)$$

Starting from the equilibrium configuration (a) fig. 11, for which each negative centre is midway between two positive centres on a line perpendicular to the slip planes, we shall give the positive lattice a displacement ξ relative to the negative lattice. Taking one of the positive centres as origin, the co-ordinates of the negative centres are $(-\xi + ma, (n + \frac{1}{2})d)$ where n and m are integral numbers positive or negative, a is the distance apart of centres of dislocation on a slip plane. If $a = d$, the positive centres lie at the corners of a square lattice. As ξ increases from zero the shear stress at the origin due to the negative centres changes, but that due to the positive centres, which remain fixed, is unaltered. If S' is the shear stress at the origin due to all the centres of dislocation (except the positive centre which is situated there), then $S' = 0$ when $\xi = 0$. The contribution to S' due to the displacement of the m th row of negative dislocations is (see (13))

$$\mu\lambda d^{-1} \{\tanh \pi d^{-1}(ma - \xi) - \tanh \pi d^{-1}ma\}. \quad (14)$$

As m increases the value of this expression rapidly decreases, so that a convergent expression is obtained for S' , namely,

$$S' = \mu\lambda d^{-1} \sum_{m=-\infty}^{+\infty} \{\tanh \pi d^{-1}(ma - \xi) - \tanh \pi d^{-1}ma\}. \quad (15)$$

If the change in S' is traced as ξ increases, it will be found that each increase in ξ by an amount a gives rise to a change in S' of amount $-2\mu\lambda d^{-1}$. This change may be regarded as being due to the separation of a layer of positive centres at the right-hand boundary of the crystal and a layer of negative centres on the left, thus it is a polarization effect arising from the distribution of dislocations on the boundary of the crystal. It will be noticed that the effect of this polarization is to increase the shear stress at all points in the field, not only at the lattice points, but an amount $-2\mu\lambda/d$ for each increase of amount a in the displacement. Thus the mean shear stress over the whole crystal due to the presence of the centres of dislocation is increased by $-2\mu\lambda/d$.

Besides the effect of the relative displacement of the two lattices of centres of dislocation the conditions of stress at the boundary of the crystal itself have an unknown effect on the stress within it. If we suppose that the mean stress in any portion of the crystal is equal to S the externally applied stress, *i.e.*, if we suppose that the mean stress taken over a portion of the crystal which includes a large number of centres of dislocation is uniform throughout the volume of the crystal and equal to S , then the mean stress due to the combined effects of the distribution of centres of dislocation and to the conditions at the boundary must be zero. Thus to obtain the true shear stress at positive centres of dislocation, due to the displacement of the lattice of negative centres, we must subtract from the expression (15) the mean value of the shear stress in the crystal due to the whole body of the centres of dislocation.

Mean Value of Shear Stress.

The mean value of the shear stress due to the distribution of centres of dislocation may be found by taking the mean value over a strip parallel to the axis of y and of breadth a . If this mean stress be represented by S_M , the change in S_M which occurs when the displacement of the negative centres relative to the positive ones (which we suppose fixed) increases by $\delta\xi$ may be regarded as being due to the removal of a strip of thickness $\delta\xi$ on one side of the strip and replacing it by a strip also of the thickness $\delta\xi$ but located in a region where the shear stress has increased by $-2\mu\lambda/d$. The mean value of S_M is therefore increased by $-2\mu\lambda\delta\xi/ad$. Hence

$$\left. \begin{aligned} \frac{dS_M}{d\xi} &= -\frac{2\mu\lambda}{ad}, \\ S_M &= -2\mu\lambda\xi/ad \end{aligned} \right\} \quad (16)$$

so that

Hence when the effects of the boundary have been taken into consideration in such a way that the mean stress throughout the body of the crystal is equal to the externally applied stress the resultant stress at a positive centre is $S + (S' - S_M)$.

If the temperature is greater than T_D so that centres of dislocation move freely under the influence of the slightest shear stress, the equilibrium condition is therefore

$$S + S' - S_M = 0, \quad (17)$$

or combining (15), (16), and (17)

$$S = \mu \lambda d^{-1} (F), \quad (18)$$

where

$$F = -\frac{2\xi}{a} + \sum_{m=-\infty}^{m=+\infty} \left\{ \tanh \frac{\pi m a}{d} - \tanh \frac{\pi}{d} (m a - \xi) \right\}. \quad (19)$$

F is a function of ξ/a and of a/d . The values shown graphically in the curves of fig. 12 have been calculated for values of ξ/a ranging from 0 to 0.5 for each of the three lattices $a = d$, $a = 0.8d$, $a = 0.5d$. It will be seen that each

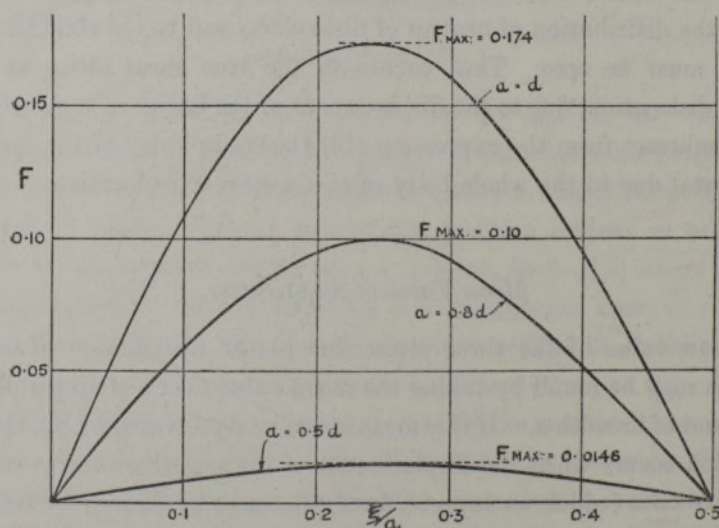


FIG. 12.

curve rises from zero at $\xi = 0$ and falls to zero at $\xi/a = 0.5$. This is to be expected because $\xi/a = 0.5$ corresponds to the equilibrium arrangement represented by (b), fig. 11. The fact that F is positive over the whole range from $\xi/a = 0$ to 0.5 shows that (b) is an unstable configuration while (a) is stable. In each case F has a maximum value (see fig. 12). For $a = d$ it is $F_{\text{max}} = 0.174$. For $a = 0.8d$ it is $F_{\text{max}} = 0.10$. For $a = 0.5d$ it is $F_{\text{max}} = 0.0146$.

It will be seen, therefore, that when the externally applied shear stress S is less than $\mu\lambda d^{-1}(F_{\max})$ a value of ξ can be found such that every centre of dislocation, positive or negative, is in a position where the shear stress is zero, so that it can remain at rest. When S rises above this value the positive centres will migrate steadily to the right and the negative ones to the left, and plastic distortion will set in. Thus the crystal has a finite shear strength

$$S = \mu\lambda d^{-1}(F_{\max}). \quad (20)$$

If the crystal is initially perfect, the first few dislocations migrate through it under the action of a shear stress which may be regarded as infinitesimally small. As the distortion proceeds, however, the number of dislocations will increase and the average value of d will decrease so that the resistance to shear will increase with the amount of distortion. This explanation of hardening by cold work is identical in principle with one put forward in a qualitative form by the present writer some years ago,* but the introduction of the idea of a unit dislocation which gives rise to a calculable field of elastic stresses has now made it possible to present the theory in a more definite and constructive form than formerly.

Diagonal Lattices of Dislocations.

So far the only arrangement of lattices which has been discussed is that in which the two kinds of dislocation lie on two interpenetrating rectangular nets. Another equilibrium arrangement is that in which the positive and negative dislocations lie on interpenetrating lattices such that each cell is a

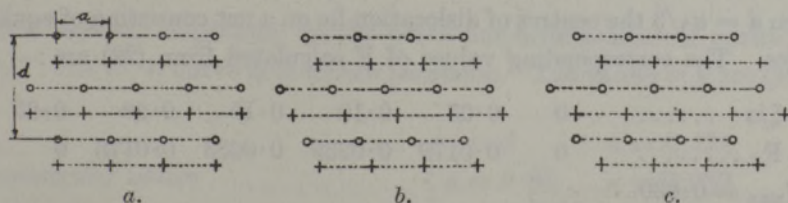


FIG. 13.—Equilibrium arrangements of dislocations, (a) and (c) stable, (b) unstable.

rhombus with one of its diagonals parallel to the slip planes. A diagonal arrangement of this kind is shown in fig. 13. If the dislocations are initially arranged as configuration (a), fig. 13, a relative displacement of the positive and negative lattices through $\frac{1}{2}a$ brings the system to configuration (c) which is the mirror image of (a) in the slip planes. Thus if (a) is stable (c) will also be stable and *vice versa*.

* 'Trans. Faraday Soc.,' *loc. cit.*

The system must also be in equilibrium when the displacement is $0.25a$ and $0.75a$. This configuration is shown in (b), fig. 13.

If a and d represent the two diagonals of the lattice cell and ξ the displacement of the positive relative to the negative lattice from the equilibrium positions (a), fig. 13, the shear stress at one of the positive centres due to a row of negative centres is (see equation (12))

$$\frac{\mu\lambda}{d} \tanh \frac{2\pi}{\lambda} \left(\frac{ma}{2} - \xi \right),$$

where m takes all integral values. Proceeding on the same lines as in the case of the rectangular lattice the condition of equilibrium is

$$S' = \mu\lambda d^{-1} (F), \quad (21)$$

where

$$F = -\frac{4\xi}{a} + \sum_{m=-\infty}^{m=+\infty} \left\{ \tanh \frac{2\pi}{d} \left(\frac{na}{2} \right) - \tanh \frac{2\pi}{d} \left(\frac{na}{2} - \xi \right) \right\}. \quad (22)$$

When $a = d$, i.e., when a square lattice is set diagonally relative to the slip planes, the following values for F were calculated from (22)

ξ/a	0	0.05	0.1	0.15	0.20	0.25
F	0	0.109	0.169	0.160	0.095	0

The maximum value is $F_{\max} = 0.17$.

The fact that F is positive in the range $\xi/a = 0$ to $\xi/a = 0.5$ shows that the equilibrium configurations (a) and (c), fig. 13, are stable while (b) (for which $\xi/a = 0.25$) is unstable.

When $d = a\sqrt{3}$ the centres of dislocation lie on a net consisting of equilateral triangles. The corresponding values of F calculated from (22) are:—

ξ/a	0	0.05	0.10	0.15	0.20	0.25
F	0	0.0179	0.0289	0.0283	0.0176	0

and $F_{\max} = 0.030$.

Relationship between Strain and the Motion of Centres of Dislocation.

In the simple case when only one centre of dislocation is present the effect of moving it across the crystal from left to right has already been seen, fig. 4. The part of the crystal above its path is displaced through a distance λ relative to the lower part. If the centre of dislocation does not travel across the crystal from side to side but merely moves through a distance x from the outer surface, the displacement is $x\lambda/L$, where L is the total length along the slip plane

through which a dislocation can move. L might, as we have so far assumed, be the total width of the specimen or it might be the width of any subdivision of the crystal the boundary of which is impervious to centres of dislocation. The total shift due to centres regularly spaced at intervals a along the slip plane is therefore $L\lambda/2a$. The shift due to a row of positive centres, which have moved into the crystal from the left, is the same as that due to a row of negative centres which have moved in from the right.

Since in the rectangular arrangement, fig. 11, the operative slip planes are spaced at distance $\frac{1}{2}d$ apart the relative shift of two slip planes separated by a thickness h is therefore $hL\lambda/ad$. By definition therefore the shear strain

$$s = L\lambda/ad. \quad (23)$$

For the diagonal arrangement of dislocations, fig. 13, the operative slip planes are situated at distance $\frac{1}{4}d$ apart so that

$$s = 2L\lambda/ad. \quad (24)$$

Theoretical Stress-strain Relationship.

We have seen that the least shear stress at which plastic flow occurs when the temperature is higher than T_D is $S = \mu\lambda d^{-1}F_{\max}$. Combining this with (23) or (24) d can be eliminated. The resulting equation is

$$\frac{S}{\mu\sqrt{s}} = \kappa \sqrt{\frac{\lambda}{L}}, \quad (25)$$

where κ is a constant depending on the particular arrangement of centres of dislocation. The (S, s) curve is therefore parabolic. The values of κ are given below.

Rectangular lattice	{	$a = d$	$\kappa = 0.174$
		$a = 0.8d$	$\kappa = 0.089$
		$a = 0.5d$	$\kappa = 0.010$
Diagonal lattice	{	$a = d$	$\kappa = 0.120$
		$a = d/\sqrt{3}$	$\kappa = 0.016.$

If the temperature is below T_D , all the foregoing work is unchanged, but the condition of equilibrium of a centre of dislocation is now

$$S - S_T = \mu\lambda d^{-1}(F),$$

where S_T is the finite stress which must be attained before a single dislocation

can migrate in a perfect crystal. The resulting strain hardening or plastic stress-strain relationship is then

$$\frac{S - S_T}{\mu \sqrt{s}} = \kappa \sqrt{\frac{\lambda}{L}}. \quad (26)$$

Choice of a Value for κ for Comparison with Experiment.

It will be seen that comparatively small variations in the arrangements of centres of dislocation produce very large changes in the value of κ . A crowding of positive centres of dislocation on one set of slip planes increases the ease with which negative centres can migrate along the intermediate planes. Thus in the rectangular lattice of dislocations a decrease in the ratio a/d corresponds with a much greater decrease in κ .

In trying to form some estimate as to which of these regular arrangements of dislocations is likely to represent most closely the true condition of a plastically strained crystal it is necessary to consider the manner in which the dislocations may arise at the boundary. Since in this theory the crystal is undisturbed by the passage of a dislocation, *i.e.*, the crystal is just as perfect behind it as it is in front, the dislocations might be supposed to arise at the boundary or within the crystal owing to thermal agitation in a random manner. The hypothesis that the points of origin of the dislocations are distributed at random precludes the formation of regular lattices of dislocations of the type we have been considering. The kind of regularity which produces very low values of κ is therefore unlikely to arise.

On the other hand, it is difficult to calculate the effect of a random production of dislocations because the mutual action of neighbouring centres of dislocation must itself tend to prevent the formation of a purely random distribution of centres inside the crystal. If, for instance, in the course of plastic distortion a positive and a negative centre approach one another from opposite sides of the crystal along planes which are separated by a very small distance h , they will not separate again till S reaches the value $\mu\lambda/2\pi h$. Thus a pseudo regularity in which the distance of a dislocation from its nearest neighbour tends always to be about $\mu\lambda/2\pi S$ is likely to arise.

We might, perhaps, visualize the whole process of plastic deformation as follows. At points distributed at random in a crystal slipping begins, owing possibly, to thermal agitation, by separation of a positive and negative dislocation. These move away from one another to an average distance apart L under the action of the external shear stress S . Slipping through one lattice

distance has therefore occurred along each line of length L which joins a positive and negative pair of centres. If at any stage there are N such pairs of dislocation lines per unit volume, the average distance of the nearest negative dislocation from any given positive one is proportional to $1/\sqrt{N}$. A very rough approximation is $1/2\sqrt{N}$.

If only the shear stress due to the nearest negative dislocation is considered, the shear stress necessary for its escape is of order of magnitude

$$S = \frac{\mu\lambda}{2\pi} (2\sqrt{N}) = \mu\lambda\sqrt{N}/\pi.$$

The strain is evidently $s = N\lambda L$ so that

$$\frac{S}{\mu\sqrt{s}} = \frac{1}{\pi} \sqrt{\frac{\lambda}{L}}. \quad (27)$$

This expression is identical with (25) if $\kappa = \pi^{-1} = 0.32$, but it is likely to overestimate S because it takes account only of the effect of the nearest dislocation of opposite sign.

In choosing a value to assume for κ in order to compare the theory with observation we might take it that there are reasons for supposing that $\kappa = 0.1$ is too low, and $\kappa = 0.3$ is too high, accordingly we may use $\kappa = 0.2$ as the most probable value in the present very imperfect state of the theory. Calculations based on this value should at any rate give the correct order of magnitude for the strength of crystals.